

Anton Matiash

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Professional Summary

Mechanical Engineering undergraduate, ranked 1st in cohort, with research experience in energy-efficient legged robots and a focus on robust sim-to-real, bio-inspired systems. Background in competitive physics (IYPT Gold).

Education

BEng in Mechanical Engineering

Sep 2024 – Jun 2027 (expected)

Swansea University

Swansea, UK

First Year Average: 89% (Top 1 in cohort), Second Year Average: 86% (Top 1 in cohort)

Selected modules: Data Analysis and Simulation (98%), Engineering Design 1 (95%), Engineering Mathematics (91%), Machines and Control (99%), AI, Statistics and Machine Learning (80%), Engineering Design 2 (81%), Dynamics I (94%)

Modern Robotics specialisation (MR)

Jun 2025 – Oct 2025

Northwestern University/Coursera

Online

Completed all 6 courses in specialisation: Foundations of Robot Motion, Robot Kinematics, Dynamics, Motion Planning and Control, Manipulation and Wheeled Mobile Robots, Capstone Project.

High School Diploma

Sep 2017 – Jun 2024

Kharkiv Physics-Mathematics Science Lyceum №27

Kharkiv, Ukraine

Last year Average: 11.1 out of 12. Top 2 in cohort

Experience

Research Intern

Jun 2026 – Sep 2026

Bristol Robotics Laboratory

Bristol, UK

- Quantified the feasibility of using Adaptive Parallel Elastic Joints (PEJs) in robotic applications by conducting a series of simulations and experiments, demonstrating their potential for high energy efficiency.
- Iterated the training pipeline through multiple setbacks to minimize Simulation-to-Real gaps as much as possible.
- Compressed findings into a scientific paper, which is in preparation for submission.

Makerspace Supervisor

Oct 2025 – Aug 2027

CREU: The Manufactory

Swansea, UK

- Helped manage a student makerspace, maintained 3D printers, laser cutters, and other equipment. Helped establish basic working procedures for the makerspace and trained new students on how to use the equipment safely.

Secretary

Jul 2026 – Jul 2027

Swansea University Table Tennis Club

Swansea, UK

- Ran club administration and organised tournaments of 30+ participants.

Honors & Awards

International Physicists' Tournament 2025

Kielce, Poland

4th of 17 countries as part of Team Ukraine: experiments, simulation and theory on Slinky staircase motion; top report score in team.

International Young Physicists' Tournament 2024 (38 countries)

Budapest, Hungary

Achieved Gold Medal as part of Team Ukraine; soap-film dynamics problem; top report score in team.

All-Ukrainian Student Physics Olympiad 2023/2024

Ivano-Frankivsk, Ukraine

3rd-place Diploma (national stage).

Selected Projects

Individual Engineering Project

Oct 2026 – Present

Robotics, Engineering, Fusion360, C++, IsaacSim/IsaacLab, Python

- Designing sprawled-leg and conventional quadrupeds of matched mass around Unitree Go actuators, to compare locomotion performance via curriculum RL in Isaac Lab, and preparing a sim-to-real-ready ROS 2 stack.
- Evaluating whether Parallel Elastic Joints reduce energy consumption more for the sprawled design.

Custom Robotics Actuator

Oct 2025 – May 2026

Robotics, Engineering, Fusion360, C++, IsaacSim, Python

- Designed, built and tested a highly modular custom robotics actuator using off-the-shelf BLDC motor and ODrive clone. Compared planetary and harmonic gearboxes and selected 7:1 planetary gearbox to achieve quasi-direct drive performance.

Capstone Project (part of the MR specialisation)

Jun 2025 – Oct 2025

MATLAB, Robotics, Computer Science

[\[Certificate Link\]](#)[\[Demonstration Video\]](#)

- Built a full kinematics, trajectory generation and control stack in MATLAB for a KUKA mobile manipulator performing pick-and-place in CoppeliaSim.

Technical Skills

Engineering: Fusion 360 (CAD & CAM), Wolfram Mathematica, MATLAB Simulink, Isaac Sim.

Programming Languages: MATLAB, Wolfram Lang, C++, C#, TypeScript, Python, SQL

Frameworks & Libraries: ROS 2, PyTorch, Isaac Lab

Languages: English (fluent), Russian (Native), Ukrainian (Native)

Other: LaTeX, Git, Docker, GitHub